Round 18 - Hermite Tree Quartic Bank (HTQB-6/3)

One carried quartic source family: quadratic path + cubic star, with sector-split memory

[Derived design] Keep the two lowest-edge connected Hermite diagrams of the degree-3 ReLU response: (i) the Q2 path with degree pattern (2,2,1,1), and (ii) the Cu1 star with degree pattern (3,1,1,1). Each birth is carried linearly for multiple layers; no n^4 tensor is materialized. The shippable truncation keeps the cubic star for 6 ages, the quadratic path for 3 ages, and applies the correction only from L8 onward. Incremental cost: 27 dense 1024^3 products + 486 n^2 scalar ops = 27.237 u.

1. Why this is the narrow family to test

- **[Quoted]** The measured Hermite family ceiling is K=2: 0.548/0.449/0.530 and K=3: 0.848/0.799/0.817 at L12/L15/L16 for h=6. K=3 follows the true-ReLU h=6 ceiling within about 0.02.
- **[Derived]** For a Gaussian restart, the leading connected degree-4 cumulant diagrams of the centered degree-3 ReLU response are exactly two tree topologies: path (2,2,1,1) from two quadratic vertices, and star (3,1,1,1) from one cubic vertex. Both have genuine four-distinct entries.
- **[Inference]** The old second-order carriers tested only the path family, so their approximately 0.44 ceiling does not kill path+star. The earlier young K4 carrier was also only 2-3 layers long, while the measured K=3 horizon requires about 5-6. This is why that kill is too narrow for HTQB.
- **[Estimated hypothesis]** Mixed Q-Cu, two-Cu and other connected K<=3 diagrams contain at least one extra correlation edge beyond these trees. In a wide random network they are the natural next corrections, but they are deliberately excluded from the first carrier because the tree pair is the cheapest possible family that contains both the required quadratic and cubic response.

2. Birth tensor, transport, and diagonal readout

[Derived] Local response coefficients. At ReLU layer b, with centered u, variance v, sigma=sqrt(v), alpha=mu/sigma:

```
L_a = Phi_a u_a,      Q_a = g_a (u_a^2-v_a),      R_a = c_a (u_a^3-3 v_a u_a)
g_a = phi(alpha_a)/(2 sigma_a),      c_a = -alpha_a phi(alpha_a)/(6 sigma_a^2).
```

[Derived] Cubic star birth (Cu1 Gaussian star). For four external indices a,b,c,d:

```
B^S_abcd = 6 sum_{hub in {a,b,c,d}} c_h prod_{j!=h} Phi_j C_hj.
```

[Derived] Quadratic path birth (Q2 Gaussian path). Sum over the six choices of quadratic slots (i,j), with remaining linear slots (k,l):

```
B^P_abcd = sum_{(i,j)} 4 g_i g_j Phi_k Phi_l C_ij (C_ik C_jl + C_il C_jk).
```

[Derived] Linear carry. A birth at post-ReLU layer b is transported to preactivation target t by the same linearized map used by the K3 source bank:

```
P_{b->t} = W[b] U[b+1] ... U[t-1],      U[r]=diag(Phi_r) W[r].
```

The carried tensor is B_{b->t}=B_b x_1 P x_2 P x_3 P x_4 P. It is never formed explicitly.

[Derived exact diagonal readout]. Let

```
X = C_b @ (Phi_b[:,None] * P),      V = (g_b[:,None] * P) * X.
star_i = 24 sum_a c_a P_ai X_ai^3,      path_i = 48 V_:i^T C_b V_:i.
```

[Measured in this session, synthetic algebra check] A direct n=4 explicit 4-index tensor contraction matched the star and path formulas to relative errors 2.07e-15 and 5.22e-16 respectively.

[Derived implementation reuse]. The public K3 bank already carries the propagator leg P and an off-diagonal covariance-response leg A. After the code's orientation convention is matched, X is the existing A leg plus the missing diagonal piece diag(v*Phi)P. Thus the star adds no dense product and no new transported dense leg. The path needs C_b only for its active ages.

3. Truncation and exact incremental cost

variant	star horizon	path horizon	target layers	dense products
ceiling carrier	6	6	L8-L16	54
shippable HTQB-6/3	6	3	L8-L16	27

[Derived] Dense products. For L8-L16 there are 9 target layers. The shippable path has 3 active births per target, hence 9x3=27 products C_b@V. Each is a real 1024x1024 by 1024x1024 product, billed as 1 u = 2^31 FLOPs. Therefore the dense part is exactly 27 u = 57,982,058,496 FLOPs.

[Derived] O(n^2) work. Using the schedule X=A+diag(v Phi)P, each star use costs exactly 7 n^2 scalar ops; each path use adds 4 n^2 scalar ops outside its GEMM. There are 54 star uses and 27 path uses, so (54x7 + 27x4)n^2 = 486 n^2 = 509,607,936 scalar FLOPs = 0.2373 u.

[Derived total] 27.2373 u incremental. State memory beyond the existing P/A source bank is three birth covariances for the live path sector: 3 x 1024^2 float32 = 12 MiB, plus six pairs of g,c vectors (<0.05 MiB).

[Estimated metered operations] About 100-250 extra ops/network if the elementwise work is stacked by active source and the three path GEMMs are issued separately per target; this is comfortably below the 1,000-op envelope. No Strassen call is required.

4. What is deliberately omitted

- **[Derived]** Q1 is zero in the Gaussian tree birth and belongs to the second-order/non-Gaussian correction family already shown to cap near 0.44 when used without cubic response.
- **[Derived]** Cu1 CxK and TxT terms, mixed Q-Cu assignments, and two-Cu terms are not in HTQB-6/3. They are higher-edge connected diagrams. If the exact tree ceiling fails, the script kills this representation before any attempt to add them.
- **[Quoted + design choice]** Corrections are used only from L8 onward because the measured teacher-pattern intervention at layers >=8 retains 86% of the total gain. This is the cost-saving truncation that makes the path sector fit the 30-u envelope.
- **[Derived]** The cubic star is kept for 6 ages; the quadratic path is kept for 3. This sector split is the only approximation in the shippable carrier beyond dropping higher-edge diagrams.

5. ONE falsifier - ceiling first

[Derived test] The accompanying script computes the exact analytic tree carrier from the packed teacher mean/C and the actual weights. It prints the project metric for (a) both tree sectors at h=6, and (b) the shippable h6-star/h3-path truncation at L12/L15/L16. It also runs an independent true-MLP Monte Carlo with N=2,000,000 by default, float32 forward and float64 accumulation, and reports the same correlations against the fresh MC K4 diagonal.

[Pre-registered gate] PASS if the untruncated h6 tree carrier reaches correlation >=0.70 at all L12/L15/L16 on the packed teacher. KILL if it is <0.55 at all three. If the ceiling passes but HTQB-6/3 drops below 0.70, increase only the path horizon before adding any higher-edge topology.

[Run status] The project data are not reachable from this execution environment, so no teacher-data correlation is labeled measured here. The self-test of the algebra did run successfully in this session.

```
python round18_htqb_falsifier.py --data kaggle/r17/data/r17_data.npz --N 2000000 --batch 4096 --backend auto
                                 --out results/summaries/round18_htqb.json
```

Script outputs: teacher correlations for ceiling_h6, ship_h6star_h3path, star_h6 and path_h6; fresh-MC correlations; packed-teacher-vs-MC sanity correlation; per-birth RMS diagnostics; final PASS/BORDERLINE/KILL verdict.

Bottom line: HTQB-6/3 is the cheapest factorised carrier I can justify from the new Round-18 evidence without re-entering a dead family. It contains both required Hermite orders, has genuine all-distinct sectors, preserves cubic-star memory for the measured six-layer horizon, and fits the derived 30-u budget. Its only serious uncertainty is representational ceiling; the supplied script resolves that before implementation work.